

Yug Dalwadi

Vadodara, Gujarat | +91 9054353956 | yugdalwadi@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

EDUCATION

Indian Institute of Technology, Jodhpur

2024–2028

B.Tech, Artificial Intelligence & Data Science

CGPA: 8.24

Expected Graduation: May 2028

EXPERIENCE

Inter IIT Tech Meet 14.0 — Real-Time Fraud Detection System [GitHub](#) | *Kafka, Pathway, River* Nov – Dec 2025

- Engineered components of a real-time fraud detection platform built on **Kafka, Pathway, and River**, processing **70k+ transactions/min** with **sub-40ms P95 pipeline latency**.
- Developed and tuned online learning models on **RiverML** using **Optuna** for real-time drift adaptation, achieving a final **F1** score of **0.92**.
- Built an LLM-powered fraud re-analysis agent using **SHAP**-based root-cause analysis report, providing natural-language explanations for flagged transactions.
- Designed an automated retraining and deployment pipeline using **BigQuery, Celery, and Weights & Biases** with health-monitoring triggers for model performance degradation.

PROJECTS

CARLA Imitation Learning for Autonomous Driving [GitHub](#) | *PyTorch, CARLA* May 2026 – Present

- Generated **1M+ multimodal driving samples** by building a CARLA data collection pipeline capturing RGB, depth, vehicle state, control, and waypoint information for imitation learning.
- Established a reproducible benchmarking framework to evaluate **6+** vision architectures, including ResNet, 3D ResNet, and R(2+1)D, for **control regression** and **waypoint prediction**.
- Analyzed large-scale driving datasets to identify waypoint generation failures, class imbalance, and collection artifacts, guiding iterative improvements to the data generation pipeline.

LLM Pre-training [GitHub](#) | *PyTorch, W&B* Mar 2026

- Implemented and pre-trained a **140M-parameter** decoder-only Transformer from scratch with RoPE, causal self-attention, and tied embeddings on **8B+ OpenWebText** tokens.
- Designed a memory-efficient data pipeline using dataset sharding and sequential streaming, enabling scalable pre-training on **8B+ OpenWebText** tokens.
- Integrated mixed-precision training, gradient accumulation, model checkpointing, and experiment tracking with **Weights & Biases** for large-scale single-GPU pre-training.

PhotoManagement [GitHub](#) | *Django, FastAPI, Pinecone* May – June 2025

- Architected an AI-powered photo management platform using **Django, FastAPI, PostgreSQL, S3, and Pinecone**, achieving **P95 retrieval latency below 720 ms** over a benchmark of **1000 indexed images**.
- Developed an image understanding pipeline using **Apple FastVLM** for caption generation, face embedding extraction for identity recognition, and automated album tagging to enrich photo metadata.
- Implemented cross-modal image retrieval by generating text captions for images, indexing their embeddings in **Pinecone**, and matching natural-language queries against visual content.

TECHNICAL SKILLS

- **Languages:** Python, C++, Go, C#
- **ML/AI:** PyTorch, Scikit-learn, River, LLM Pre-training, Vector Search, Hugging Face, NumPy
- **Systems:** Kafka, Docker, FastAPI, Django, PostgreSQL, Neo4j, Apache Arrow, Celery, Pinecone
- **Tools:** Git, Linux, Weights & Biases, Optuna, BigQuery

COURSEWORK & ACTIVITIES

- **Relevant Coursework:** Pattern Recognition and Machine Learning, Probability Statistics & Stochastic Processes, Linear Algebra, OS and Networking, Computer Architecture and Compilers.
- **Leadership & Activities:** Core Team Member, Devlup (Open Source Club) and RAID (AI Club); mentored 6 students during WARP'26 on a depth-conditioned image generation project using StyleGAN; guided another cohort of 6 students during SR'26 in developing a graph-based research paper learning path generation tool.
- **Technical Writing:** Authored [blogs](#) on machine learning, deep learning, and AI systems.